

EE1060: Differential Equations and Transform Techniques**Practice Problem Set - 07**

Practice problems in topics covered till: Lecture 28

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Compute the Fourier Transform of the signal $x(t)$ given by (1)

$$x(t) = \begin{cases} A & |t| \leq \frac{\epsilon}{2} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The signal $x(t)$ is often represented as $A \text{ rect}\left(\frac{t}{\epsilon}\right)$, where $\text{rect}(x)$ denotes a rectangular function, typically defined as

$$\text{rect}(t) = \begin{cases} 1 & |t| \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

2. Compute the Fourier Transform of the sinc function given by (3).

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t} \quad (3)$$

Hint: Use the duality property.

3. Compute the Fourier Transform of the scaled sinc function $x(t)$ given by (4).

$$x(t) = \frac{\sin(\omega_0 t)}{\omega_0 t} \quad (4)$$

4. Compute the Fourier Transform of the triangular wave $x(t)$ given by (5).

$$x(t) = \begin{cases} A \left(1 - \frac{|t|}{\epsilon}\right) & |t| \leq \epsilon \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

5. Compute the Fourier Transform of the ramp function given by (6).

$$\text{sinc}(t) = At \quad (6)$$

6. Show that the Fourier Transform of the [Gaussian signal](#)

$$x(t) = e^{-2t^2} \text{ where } \alpha > 0 \quad (7)$$

is also a Gaussian function given by

$$X(\omega) = \sqrt{\frac{\pi}{2}} e^{-\frac{\omega^2}{8}} \quad (8)$$

7. Compute the Fourier Transform of the signal $x(t)$ given by (9).

$$x(t) = \sin(\omega_0 t - \theta) \quad (9)$$

8. ** Show that the Fourier Transform of the signal $x(t) = \text{sinc}^2(t)$ is

$$X(\omega) = \text{tri}\left(\frac{\omega}{\pi}\right) \quad (10)$$

where

$$\text{tri}\left(\frac{t-a}{b}\right) = \begin{cases} 1 - \frac{|t-a|}{b} & |t-a| \leq b \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

Hint: Use properties of FT to derive this result.

9. Compute the Fourier Transform of $x(t)$ given by

$$x(t) = e^{\alpha t} u(-t) \text{ where } \alpha > 0 \quad (12)$$

10. Consider a system whose input and output are related by the first-order ODE

$$\frac{dy}{dt} + ay = bx(t) \quad (13)$$

- (a) Show the block diagram representation of the system.
 - (b) Compute the impulse response of the system. Comment on the stability of the system based on the impulse response.
 - (c) Compute the step response of the system. Compute the rise time and settling time of the system.
11. Consider a system whose input and output are related by the first-order ODE

$$\frac{dy}{dt} + ay = bx(t) \quad (14)$$

- (a) Show the block diagram representation of the system.
- (b) Compute the impulse response of the system. Comment on the stability of the system based on the impulse response.

- (c) Compute the step response of the system. Compute the rise time and settling time of the system.

12. ** Consider a system whose input and output are related by the second-order ODE

$$\frac{dy}{dt} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2 y = x(t) \quad (15)$$

- (a) Show the block diagram representation of the system.
- (b) Compute the impulse response of the system when $|\zeta| < 1$. Comment on the stability of the system based on the impulse response.
- (c) Compute the step response of the system when $|\zeta| < 1$. Compute the rise time and settling time of the system.

13. Determine if the system whose input-output relationship is given by (16) is linear and time-invariant.

$$\frac{dy}{dt} - \sin(y(t)) = x(t) \quad (16)$$

14. Determine if the system whose input-output relationship is given by (17) is linear and time-invariant.

$$\frac{dy}{dt} = x(t - 2) \quad (17)$$

15. Determine if the system whose input-output relationship is given by (18) is linear and time-invariant.

$$\frac{dy}{dt} = x(t) + \cos(t) \quad (18)$$

16. Determine if the system whose input-output relationship is given by (19) is linear and time-invariant.

$$y(t) = 3x^2(t) + 2x(t) \quad (19)$$

17. Determine if the system whose input-output relationship is given by (20) is linear and time-invariant.

$$y(t) = tx(t) \quad (20)$$

Please Note: Some of the problems presented in this document have been adapted from textbooks referenced for this course. These problems are included to help students practice and understand key concepts. The material is used for educational purposes only. We have made every effort to provide proper citations, but if any are inadvertently omitted, it was not intentional. All copyright and trademark rights belong to the original authors and publishers.